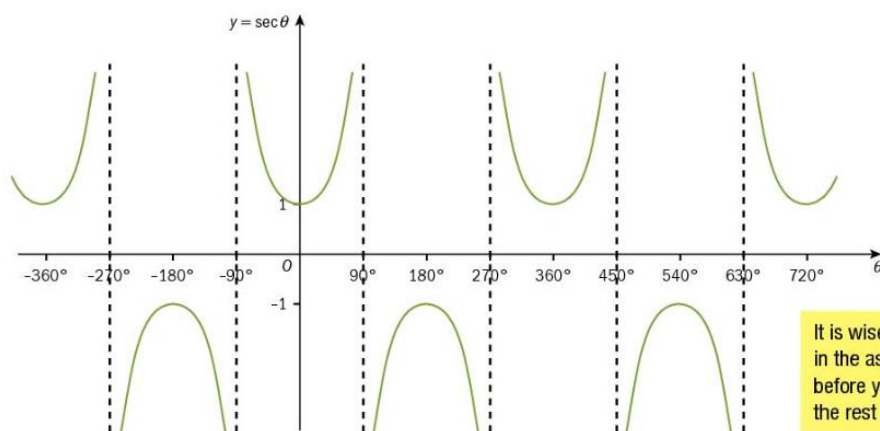


1. Trigonometry Spire — Trigonometry
a. Secant, cosecant, cotangent functions

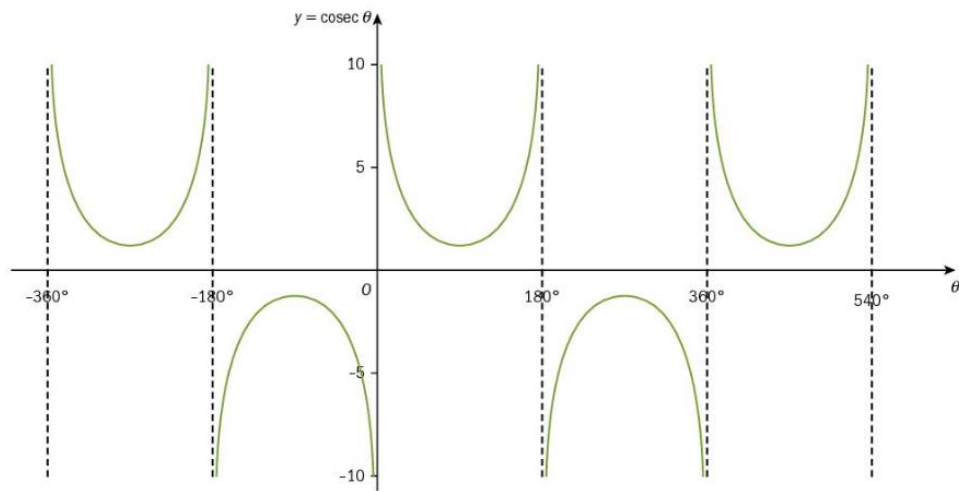
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

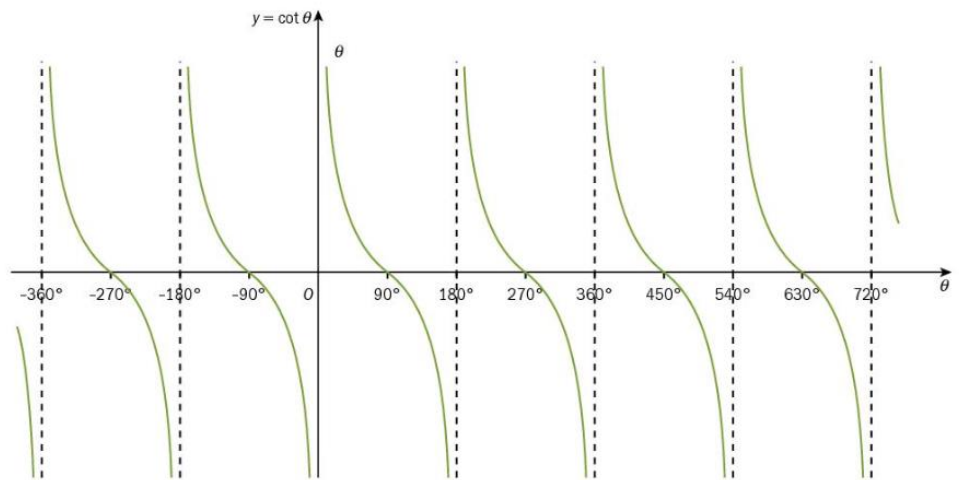
$$\cot \theta = \frac{1}{\tan \theta} = \left(\frac{\cos \theta}{\sin \theta} \right)$$



It is wise to mark
in the asymptotes
before you sketch
the rest of the graph.



The graph of $y = \cot \theta$ can be obtained from the graph of $y = \tan \theta$.



6. Solve each of these equations for $0 \leq \theta \leq 2\pi$. Give your answers in exact form.

a) $\operatorname{cosec} \theta = 1$ b) $\sec \theta = 1$ c) $\cot \theta = -1$
 d) $\sec \theta = \frac{2}{\sqrt{3}}$ e) $\cot \theta = 0$ f) $\operatorname{cosec} \theta = -\frac{2}{\sqrt{3}}$

7. Solve each of these equations for $0^\circ \leq x < 360^\circ$. Give your answers correct to 1 decimal place.

a) $\sec x = 3$ b) $\operatorname{cosec} x = 4$ c) $\cot x = 0.9$
 d) $\cot x = 5$ e) $\operatorname{cosec} x = -\frac{9}{7}$ f) $\sec x = -\sqrt{6}$

8. a) Solve the equation $\sec x = 7$ for $0^\circ < x < 360^\circ$.

b) Solve the equation $\sec^2 x = 49$ for $0^\circ < x < 360^\circ$.

9. Solve the equation $8 \cot x - 5 = 7$ for $0^\circ < x < 360^\circ$.

10. Solve the equation $7 + 2 \operatorname{cosec} \theta = 4$ for $0 < \theta < 2\pi$.

b. Pythagorean identities expanded

$$1 + \tan^2 \theta \equiv \sec^2 \theta$$

$$1 + \cot^2 \theta \equiv \operatorname{cosec}^2 \theta$$

Example 3

Prove the identity $(\tan \theta + \cot \theta)^2 \equiv \sec^2 \theta + \operatorname{cosec}^2 \theta$.

$$(\tan \theta + \cot \theta)^2 \equiv \tan^2 \theta + 2 \tan \theta \cot \theta + \cot^2 \theta$$

Start with the left-hand side and expand the brackets.

$$\equiv \tan^2 \theta + 2 \tan \theta \frac{1}{\tan \theta} + \cot^2 \theta$$

Replace $\cot \theta$ with $\frac{1}{\tan \theta}$.

$$\equiv \tan^2 \theta + 2 + \cot^2 \theta$$

$$\equiv \sec^2 \theta - 1 + 2 + \operatorname{cosec}^2 \theta - 1$$

Use the identities to replace $\tan^2 \theta$ with $\sec^2 \theta - 1$ and $\cot^2 \theta$ with $\operatorname{cosec}^2 \theta - 1$ and then simplify.

$$\equiv \sec^2 \theta + \operatorname{cosec}^2 \theta$$

Example 4

Solve the equation $\sec^2 \theta + \tan \theta - 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$.

$$1 + \tan^2 \theta + \tan \theta - 1 = 0$$

Substitute $1 + \tan^2 \theta$ for $\sec^2 \theta$ to get a quadratic equation in $\tan \theta$.

$$\tan^2 \theta + \tan \theta = 0$$

$$\tan \theta (\tan \theta + 1) = 0$$

Factorise the equation and solve for $\tan \theta$.

$$\tan \theta = 0 \text{ or } \tan \theta + 1 = 0$$

$$\tan \theta = 0 \text{ or } \tan \theta = -1$$

$$\theta = 0^\circ, 135^\circ, 180^\circ, 315^\circ, 360^\circ$$

4. Solve the equation $\tan^2 \theta - \sec \theta - 5 = 0$ for $0^\circ \leq \theta \leq 360^\circ$.

5. Prove the identity $\operatorname{cosec} \theta + \cot \theta \equiv \frac{1}{\operatorname{cosec} \theta - \cot \theta}$.

Hint: Start with the right hand side and consider the denominator.

6. Prove the identity $\frac{\tan \theta}{1 + \sec \theta} + \frac{1 + \sec \theta}{\tan \theta} \equiv \frac{2}{\sin \theta}$.

7. Prove the identity $(\sec \theta - \tan \theta)^2 \equiv \frac{1 - \sin \theta}{1 + \sin \theta}$.

8. Prove the identity $\frac{\cot \theta}{1 + \cot^2 \theta} \equiv \sin \theta \cos \theta$.

9. Prove the identity $\sec^2 \theta + \operatorname{cosec}^2 \theta \equiv \sec^2 \theta \operatorname{cosec}^2 \theta$.

10. Solve the equation $4 \cot^2 \theta - 2 \cot \theta = 3 \operatorname{cosec}^2 \theta$ for $0^\circ \leq \theta \leq 360^\circ$.

11. Solve the equation $\sec \theta = 3 \cos \theta + 1$ for $0^\circ \leq \theta \leq 360^\circ$.

c. Trig addition formulas

$$\sin(A + B) \equiv \sin A \cos B + \cos A \sin B$$

$$\sin(A - B) \equiv \sin A \cos B - \cos A \sin B$$

$$\cos(A + B) \equiv \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) \equiv \cos A \cos B + \sin A \sin B$$

$$\tan(A + B) \equiv \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A - B) \equiv \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Example 7

Solve the equation $\cos(\theta + 60^\circ) = 2 \sin(\theta - 45^\circ)$ for $0^\circ \leq \theta \leq 360^\circ$.

$$\cos \theta \cos 60^\circ - \sin \theta \sin 60^\circ = 2(\sin \theta \cos 45^\circ - \cos \theta \sin 45^\circ)$$

Expand both expressions.

$$\frac{1}{2} \cos \theta - \frac{\sqrt{3}}{2} \sin \theta = 2 \left(\frac{1}{\sqrt{2}} \sin \theta - \frac{1}{\sqrt{2}} \cos \theta \right)$$

Substitute values for $\sin 45^\circ$, $\cos 45^\circ$, $\sin 60^\circ$ and $\cos 60^\circ$.

$$\left(\frac{1}{2} + \frac{2}{\sqrt{2}} \right) \cos \theta = \left(\frac{2}{\sqrt{2}} + \frac{\sqrt{3}}{2} \right) \sin \theta$$

Collect terms in $\sin \theta$ and $\cos \theta$.

$$\frac{\sqrt{2} + 4}{2\sqrt{2}} \cos \theta = \frac{4 + \sqrt{3}\sqrt{2}}{2\sqrt{2}} \sin \theta$$

$$\frac{\sqrt{2} + 4}{4 + \sqrt{3}\sqrt{2}} = \tan \theta$$

Rearrange to find $\tan \theta$.

$$\tan \theta = 0.8394 \dots$$

Find all the angles.

$$\theta = 40.0^\circ, 220.0^\circ$$

We have worked in exact terms until the final step, but you may wish to use a calculator at an earlier stage in the solution.

5. Evaluate, in exact form,

a) $\cos 10^\circ \cos 20^\circ - \sin 10^\circ \sin 20^\circ$

b) $\sin 75^\circ \cos 45^\circ + \cos 75^\circ \sin 45^\circ$

c) $\frac{\tan 103^\circ - \tan 58^\circ}{1 + \tan 103^\circ \tan 58^\circ}$

d) $\frac{\tan 75^\circ + 1}{\tan 75^\circ - 1}$

7

6. Simplify

a) $\sin(\alpha + \beta) + \sin(\alpha - \beta)$

b) $\cos(\alpha + \beta) + \cos(\alpha - \beta)$

c) $\sin(\alpha - \beta) \cos \alpha - \cos(\alpha - \beta) \sin \alpha$

7. Use the addition formulae to show that

a) $\sin\left(\theta + \frac{\pi}{2}\right) \equiv \cos \theta$

b) $\sin(90^\circ - \theta) \equiv \cos \theta$

c) $\cos(90^\circ - \theta) \equiv \sin \theta$

d) $\cos(180^\circ + \theta) \equiv -\cos \theta$

e) $\tan(\theta + 180^\circ) \equiv \tan \theta$

f) $\tan(\pi - \theta) \equiv -\tan \theta$

8. Prove the identity $\operatorname{cosec}(\theta + \varphi) \equiv \frac{\operatorname{cosec} \theta \operatorname{cosec} \varphi}{\cot \theta + \cot \varphi}$.

d. Trig double angle formulas

$$\sin 2A \equiv 2 \sin A \cos A$$

$$\begin{aligned}\cos 2A &\equiv \cos^2 A - \sin^2 A \\ &\equiv 2 \cos^2 A - 1 \\ &\equiv 1 - 2 \sin^2 A\end{aligned}$$

$$\tan 2A \equiv \frac{2 \tan A}{1 - \tan^2 A}$$

Example 8

Solve the following equations for $0^\circ \leq \theta \leq 360^\circ$.

a) $4 \sin 2\theta = \sin \theta$ b) $\cos 2\theta = \cos \theta$

▶ Continued on the next page

Trigonometry

a) $4 \times 2 \sin \theta \cos \theta = \sin \theta$

Write the equation in terms of $\sin \theta$ and $\cos \theta$ only.

$8 \sin \theta \cos \theta - \sin \theta = 0$

Write in the form $f(\theta) = 0$.

$\sin \theta (8 \cos \theta - 1) = 0$

Factorise the LHS and solve the equation.

$\sin \theta = 0$ or $\cos \theta = \frac{1}{8}$

$\theta = 0^\circ, 180^\circ, 360^\circ$ or $\theta = 82.8^\circ, 360^\circ - 82.8^\circ$

$\theta = 0^\circ, 82.8^\circ, 180^\circ, 277.2^\circ, 360^\circ$

b) $2 \cos^2 \theta - 1 = \cos \theta$

Write the equation in terms of $\cos \theta$ only.

$2 \cos^2 \theta - \cos \theta - 1 = 0$

$(2 \cos \theta + 1)(\cos \theta - 1) = 0$

Factorise and solve the resulting quadratic equation in $\cos \theta$.

$\cos \theta = -\frac{1}{2}$ or $\cos \theta = 1$

$\theta = 120^\circ, 240^\circ$ or $\theta = 0^\circ, 360^\circ$

$\theta = 0^\circ, 120^\circ, 240^\circ, 360^\circ$

Example 9

Express $\cos 3\theta$ in terms of $\cos \theta$.

$\cos 3\theta \equiv \cos(2\theta + \theta)$

Express $\cos 3\theta$ in a way that the addition formulae may be used.

$\equiv \cos 2\theta \cos \theta - \sin 2\theta \sin \theta$

Use the double angle formulae.

$\equiv (2 \cos^2 \theta - 1) \times \cos \theta - 2 \sin \theta \cos \theta \times \sin \theta$

$\equiv 2 \cos^3 \theta - \cos \theta - 2 \sin^2 \theta \cos \theta$

$\equiv 2 \cos^3 \theta - \cos \theta - 2(1 - \cos^2 \theta) \cos \theta$

Ensure that the expression is expressed in terms of $\cos \theta$ only.

$\equiv 4 \cos^3 \theta - 3 \cos \theta$

5. If $\tan \theta = 3$, where $180^\circ \leq \theta \leq 270^\circ$, find the exact values of
 a) $\sin \theta$ b) $\cos \theta$ c) $\tan 2\theta$ d) $\cos 2\theta$.
6. a) Express $(\sin x - \cos x)^2$ in terms of $\sin 2x$.
 b) Express $(\cos^4 x - \sin^4 x)(\cos^2 x - \sin^2 x)$ in terms of $\cos 2x$.
7. Express $\sin 4\theta \sin \theta$ in terms of $\cos \theta$.
- 8 a) i) By considering $\sin(2\theta + \theta)$, express $\sin 3\theta$ in terms of $\sin \theta$.
 ii) Solve the equation $\sin 3\theta = \sin \theta$ for $0^\circ \leq \theta \leq 360^\circ$.
 b) Express $\tan 3\theta$ in terms of $\tan \theta$.
9. Express $\cot 2\theta$ in terms of $\cot \theta$.
10. Prove that a) $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$ b) $\frac{\sin 2\theta}{1 - \cos 2\theta} = \cot \theta$.

e. Polar form with trig

$$a \sin \theta + b \cos \theta \equiv R \sin(\theta + \alpha), \quad \text{where } R = \sqrt{a^2 + b^2}$$

$$\text{and } \alpha = \tan^{-1} \frac{b}{a}$$

Example 10

- a) Express $2 \sin \theta + \cos \theta$ in the form $R \sin(\theta + \alpha)$ where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$, giving the exact value of R and the value of α correct to 2 decimal places.
 b) Hence solve the equation $2 \sin \theta + \cos \theta = 2$ for $0^\circ \leq \theta \leq 360^\circ$.

a) $2 \sin \theta + \cos \theta \equiv R \sin(\theta + \alpha)$

$$\equiv R \sin \theta \cos \alpha + R \cos \theta \sin \alpha$$

Expand the expression.

$$R \cos \alpha = 2$$

Compare the coefficients of $\sin \theta$ and $\cos \theta$.

$$R \sin \alpha = 1$$

$$R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 2^2 + 1^2$$

Square and add to find the exact value of R .

$$R^2 (\cos^2 \alpha + \sin^2 \alpha) = 5$$

$$R = \sqrt{5}$$

Also,

$$\frac{R \sin \alpha}{R \cos \alpha} = \frac{1}{2}$$

Divide to find $\tan \alpha$ and then find α to the required accuracy.

$$\tan \alpha = \frac{1}{2}$$

$$\alpha = \tan^{-1} \frac{1}{2} = 26.57^\circ$$

$$\text{Thus, } 2 \sin \theta + \cos \theta = \sqrt{5} \sin(\theta + 26.57^\circ)$$

b) $2 \sin \theta + \cos \theta = 2$

$$\sqrt{5} \sin(\theta + 26.57^\circ) = 2$$

Substitute $\sqrt{5} \sin(\theta + 26.57^\circ)$ for $2 \sin \theta + \cos \theta$.

$$\sin(\theta + 26.57^\circ) = \frac{2}{\sqrt{5}}$$

$$\theta + 26.57^\circ = \sin^{-1} \frac{2}{\sqrt{5}}$$

Divide both sides by $\sqrt{5}$.

$$\theta + 26.57^\circ = 63.43^\circ, 180^\circ - 63.43^\circ, 360^\circ + 63.43^\circ$$

Find all the values for $\theta + 26.57^\circ$.

$$\theta = 36.9^\circ, 90^\circ, 396.9^\circ$$

$$\theta = 36.9^\circ, 90^\circ \text{ for } 0^\circ \leq \theta \leq 360^\circ$$

Subtract 26.57° from each value and check you only include values in the required interval.

Example 11

- a) Express $3 \cos \theta + \sqrt{3} \sin \theta$ in the form $R \cos(\theta - \alpha)$ where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$, stating the exact values of R and α .
- b) **Determine** the greatest and least possible values of $[(3 \cos \theta + \sqrt{3} \sin \theta)^2 - 5]$ as θ varies.

a) $3 \cos \theta + \sqrt{3} \sin \theta = R \cos(\theta - \alpha)$

$$\equiv R \cos \theta \cos \alpha + R \sin \theta \sin \alpha \quad \leftarrow \text{Expand the expression.}$$

$$R \cos \alpha = 3$$

$$R \sin \alpha = \sqrt{3}$$

$\leftarrow$ Equate coefficients of $\sin \theta$ and $\cos \theta$.

$$R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 3^2 + (\sqrt{3})^2$$

$$R = \sqrt{12} \text{ (or } 2\sqrt{3}\text{)}$$

Also,

$$\tan \alpha = \frac{\sqrt{3}}{3}$$

$$\alpha = 30^\circ$$

$$3 \cos \theta + \sqrt{3} \sin \theta = \sqrt{12} \cos(\theta - 30^\circ)$$

$\leftarrow$ Write out the alternative form of the expression.

b) $3 \cos \theta + \sqrt{3} \sin \theta = \sqrt{12} \cos(\theta - 30^\circ)$

$\leftarrow$ Use the expression found in part (a).

$$\text{but } -1 \leq \cos(\theta - 30^\circ) \leq 1$$

$$\text{so } -\sqrt{12} \leq \sqrt{12} \cos(\theta - 30^\circ) \leq \sqrt{12}$$

$\leftarrow$ Use the fact that the cosine of an angle of any size is between -1 and 1 .

$$-\sqrt{12} \leq (3 \cos \theta + \sqrt{3} \sin \theta) \leq \sqrt{12}$$

$$0 \leq (3 \cos \theta + \sqrt{3} \sin \theta)^2 \leq 12$$

$\leftarrow$ Remember that the square of an expression cannot be negative.

$$-5 \leq (3 \cos \theta + \sqrt{3} \sin \theta)^2 - 5 \leq 7$$

Greatest value of $[(3 \cos \theta + \sqrt{3} \sin \theta)^2 - 5]$ is 7

Least value of $[(3 \cos \theta + \sqrt{3} \sin \theta)^2 - 5]$ is -5

4. a) Express each of the following in the form $R \cos(\theta - \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$, giving the exact value of R and the value of α correct to 2 decimal places.
- i) $3 \cos \theta + 4 \sin \theta$ ii) $5 \cos \theta + 12 \sin \theta$
 iii) $\sqrt{2} \cos \theta + \sin \theta$ iv) $\sin \theta + \cos \theta$
- b) Determine the greatest and least possible values of each of the expressions in part (a).
- c) For each of the expressions in part (a), find a value of θ for which the expression has its greatest value.
5. a) Express $\sin \theta + 2 \cos \theta$ in the form $R \sin(\theta + \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$, giving the exact value of R and the value of α correct to 2 decimal places.
- b) Hence solve the equation $\sin \theta + 2 \cos \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$.
6. a) Express $\sqrt{3} \sin \theta - \cos \theta$ in the form $R \sin(\theta - \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$, giving the exact values of R and α .
- b) Hence solve the equation $\sqrt{3} \sin \theta - \cos \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$.
- c) Find the greatest and least possible values of $(\sqrt{3} \sin \theta - \cos \theta)^2$ as θ varies.

